

GraphLIME on RDF Knowledge Graphs

Explaining R-GCN entity classification in the vocabulary of the graph

XAI Laboratory Course — Project Report

Outline

1. The problem
2. The proposed design
3. Method
4. Experimental setup
5. Results
6. Limitations and conclusions

Motivation: an R-GCN is opaque twice over

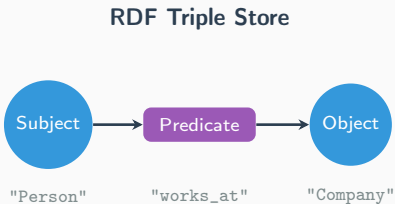
Relational Graph Convolutional Networks are the standard baseline for entity classification on RDF knowledge graphs. They are unreadable at two distinct levels:

1. **As a GNN.** The decision is buried in message passing.
2. **As an RDF model.** Its usual input is a **one-hot entity id**, so even a perfect feature-attribution method could only report “*entity #4211 mattered*”.

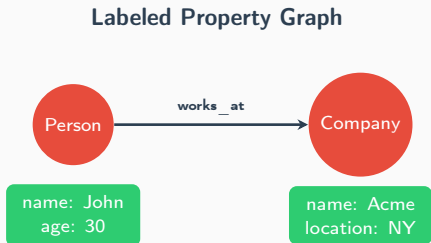
The question of this project

Can we make the explanation *speak RDF*? An explanation for “person X belongs to research group Y” should read “*because X carries `swrc:publication` edges toward papers of that group*” — predicates and terms of the graph, not tensor indices.

Why the data model matters: RDF vs. Labeled Property Graph



Nodes carry *no* properties.
All information is on the typed edge.



Nodes carry key-value properties:
a feature vector already exists.

Feature-attribution explainers silently assume the *right-hand* model. RDF is the left-hand one.

The crux: GraphLIME cannot be applied to RDF as published

GraphLIME's entire mechanism operates on a node **feature matrix**: it collects the feature vectors of the target's neighbours and asks, via kernels, which *feature dimensions* the model's outputs depend on.

Designed for citation networks

- every node has a bag-of-words vector;
- every dimension *is* a word — readable by construction.

In the RDF setting

- entities have *no feature vectors at all*;
- the one-hot entity id offers no dimension a human could read.

Consequence

With no feature matrix there is nothing for HSIC Lasso to select from; with anonymous embedding dimensions any selection would be unreadable anyway. The explainer is not merely weakened — **its input is undefined**. Defining that input *is* the project.

Manufacturing the missing feature space

We build the feature space **from the graph itself**, in the graph's own vocabulary — the methodological contribution of this project.

Space A — predicate counts

out:<p> counts the node's outgoing <p> edges, in:<p> its incoming ones.

Coarse, dense, stable. ≈ 90 columns on AIFB.

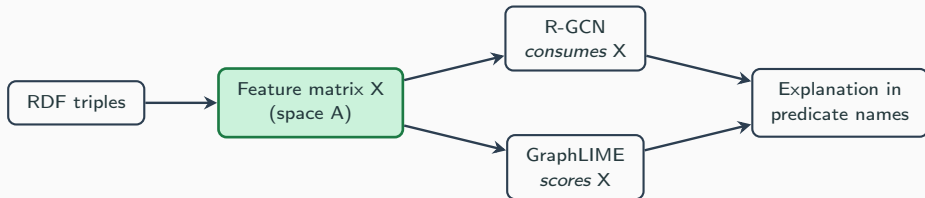
Space B — predicate–object pairs

out:<p>=<o> marks having <p> toward the *specific* term <o>.

Expressive but high-dimensional and sparse, which starves the kernel estimates; pruned by a minimum-support threshold.

Comparing A against B is itself one of the reported experiments.

The decisive move: one shared vocabulary



The R-GCN consumes space A **in place of** the one-hot input. Model and explainer therefore share one vocabulary, which buys two things:

1. every feature the explainer scores is a readable statement about the graph (“has publication edges”, “is author of ...”);
2. **fidelity becomes well-defined** — otherwise undefined, since the explainer’s features would not be what the model consumes — as masking feature columns of the actual model input.

The model: R-GCN

Graph convolution extended to multi-relational graphs by giving every relation its own transformation:

$$h_i^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} W_r^{(l)} h_j^{(l)} + W_0^{(l)} h_i^{(l)} \right)$$

with $\mathcal{N}_i^r$ the neighbours reached through relation r , $c_{i,r} = |\mathcal{N}_i^r|$, and W_0 a self-loop term.

Why per-relation weights. In a knowledge graph the meaning lives in the edge *types*; a plain GCN would collapse `author` and `employs` into an undifferentiated “neighbour”.

Basis decomposition. One matrix per relation is prohibitive (45 relations on AIFB), so $W_r^{(l)} = \sum_{b=1}^B a_{rb}^{(l)} V_b^{(l)}$ — parameter sharing that doubles as regularisation.

HSIC: measuring non-linear dependence

The Hilbert–Schmidt Independence Criterion is a kernel measure of statistical dependence. With kernel matrices K , L and centering matrix $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$:

$$\widehat{\text{HSIC}}(K, L) = \frac{\text{tr}(KHLH)}{(n-1)^2}$$

For characteristic kernels (RBF, median-heuristic bandwidth) it vanishes exactly when the variables are independent.

Why kernels and not a linear surrogate

$y = x^2$ with x symmetric around zero has Pearson correlation ≈ 0 but clearly positive HSIC. A GNN's output is a non-linear function of its input, so a linear surrogate would measure the wrong thing. Our property tests assert this explicitly.

HSIC Lasso and its three-term reading

With centered, Frobenius-normalised kernels $\bar{K}_k$ (features) and $\bar{L}$ (model output), solve the non-negative Lasso

$$\min_{\beta \geq 0} \frac{1}{2} \left\| \bar{L} - \sum_{k=1}^d \beta_k \bar{K}_k \right\|_F^2 + \rho \|\beta\|_1.$$

Expanding the square exposes the **explainable three-term form**:

$$\min_{\beta \geq 0} \underbrace{- \sum_k \beta_k \text{HSIC}(x_k, y)}_{\text{relevance}} + \underbrace{\frac{1}{2} \sum_{k,l} \beta_k \beta_l \text{HSIC}(x_k, x_l)}_{\text{redundancy}} + \underbrace{\rho \|\beta\|_1}_{\text{sparsity}}$$

Term 1 rewards features the output depends on; term 2 penalises features redundant with one another; term 3 forces most weights to zero. The resulting $\beta_k \geq 0$ is feature k 's importance.

Unlike LIME, GraphLIME **perturbs nothing**. To explain node v :

- the “samples” are the **real nodes of v ’s h -hop neighbourhood** ($h = 2$ here);
- X collects their features, Y the model’s predicted probability vectors;
- HSIC Lasso on (X, Y) yields a sparse β over feature dimensions.

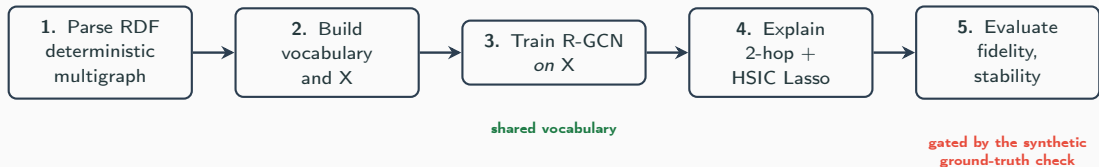
It explains features, not edges

On a graph whose nodes have no features the method has no input at all — the situation of Section 1.

Its samples are a neighbourhood

A small neighbourhood means few samples and a meaningless HSIC estimate, so the pipeline **refuses** nodes below a minimum size rather than emit noise.

The pipeline end to end



Non-zero β are mapped back to **predicate names**. HSIC and the solver are implemented from scratch, with property tests written *before* the implementation: symmetry, non-negativity, $\text{nHSIC} \in [0, 1]$, joint-permutation invariance, weight-splitting for duplicated features, and sparsity monotone in ρ .

GNNExplainer (Ying et al., 2019) natively optimises a soft mask over *edges* via mutual information.

A structural incompatibility

Its edge masks cannot be applied to RGCNConv, which runs one propagate per relation on edge subsets — a full-graph mask can never align.

Resolution. The baseline uses GNNExplainer's **attribute mask** over the same feature matrix: it scores exactly the features GraphLIME scores, making the two rankings directly comparable at the predicate level. The price is that the comparison covers feature attribution only.

Datasets and model

Dataset	Nodes	Triples	Predicates	Task
AIFB	8,285	29,043	45	research-group affiliation (4 classes)
MUTAG	23,644	74,227	23	mutagenicity (2 classes)

Model

- 2-layer RGCNConv, 30 bases, hidden 32
- Adam, lr 0.01, 200 epochs, 5 seeds
- predicate-count features, both directions

Explanations

- 2-hop neighbourhoods, subsampled above 200 nodes
- RBF kernels, median heuristic, $\rho = 0.1$
- neighbourhoods under 10 nodes refused

Label-revealing predicates (`employs/affiliation`, `isMutagenic`) are absent from the dumps; a blocklist **asserts** their absence at every load.

The correctness gate: synthetic ground truth

On real data there is no ground-truth explanation. Before *any* number on AIFB or MUTAG was produced, the pipeline had to recover a planted rule.

Generated graph where class 1 $\iff$ the entity has predicate p_7

- GraphLIME ranked out:syn:p7 **first** for $\geq 95\%$ of explained nodes at 0% noise;
- it kept the target **top-3** under two distractor predicates 90%-correlated with it;
- $\text{fid}^+ > \text{random-}k$ is *asserted* here.

It passed on the first run.

On real data the numbers are **reported, not asserted**: no threshold is ever lowered to make progress.

The evaluation metrics (1/2)

Let S_k be the top- k features by β , and $f_{\hat{y}}$ the model's probability for the predicted class.

Fidelity+ — necessity (*higher is better*)

$$\text{fid}^+ = \frac{1}{|V|} \sum_{v \in V} [f_{\hat{y}}(x_v) - f_{\hat{y}}(x_v \setminus S_k)]$$

Zero the top- k columns: if they matter, the prediction should fall.

Fidelity- — sufficiency (*lower is better*)

$$\text{fid}^- = \frac{1}{|V|} \sum_{v \in V} [f_{\hat{y}}(x_v) - f_{\hat{y}}(x_v^{S_k})]$$

Keep *only* the top- k : if they suffice, the prediction should survive.

Random- k control. Masking any k columns perturbs the prediction, so both metrics are recomputed on a random k -subset. The evidence is the **gap**, not the raw value.

The evaluation metrics (2/2)

Stability — Jaccard@5

$J(S, S') = \frac{|S \cap S'|}{|S \cup S'|} \in [0, 1]$, averaged over explained nodes, computed across **training seeds** and across **hops** $\in \{1, 2, 3\}$. An explainer that changes its answer when nothing meaningful changed is not trustworthy.

Agreement — Jaccard@5 over predicates

The same index after collapsing each feature to its predicate URI, used for **space A vs. space B** and **GraphLIME vs. GNNExplainer**. Not a correctness measure — there is no reference ranking — so it is read as a sensitivity analysis.

Refusal rate

Fraction of entities whose neighbourhood falls below `min_neighborhood = 10`. A refusal is a finding about applicability, not a failure.

Classification accuracy

Dataset	Test accuracy (5 seeds)	Best seed	R-GCN paper (one-hot)
AIFB	0.8833 ± 0.0534	0.9444	0.958
MUTAG	0.7559 ± 0.0305	0.7941	0.732

Interpretation. The interpretable feature space is no handicap on MUTAG, whose 5-seed mean *exceeds* the one-hot R-GCN, while AIFB lands several points below its 0.958. AIFB's classes hinge on *which* specific entity a person is connected to — exactly what per-entity embeddings encode for free and predicate counts deliberately discard.

The cost of interpretability is visible, dataset-dependent, and paid in the input representation
— not in the explainer.

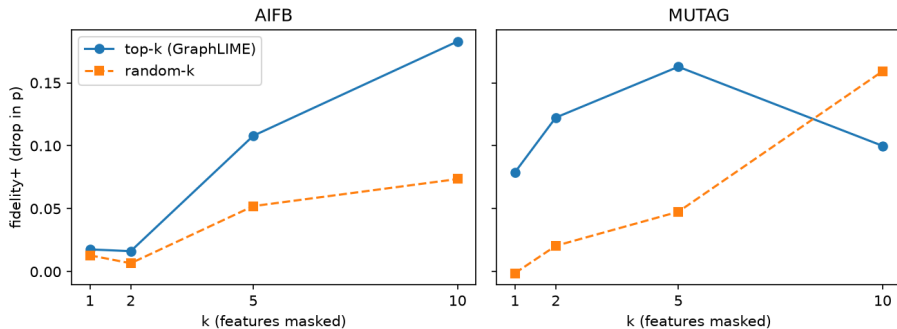
Explanation fidelity: the numbers

Dataset	k	fidelity+	random+	fidelity−	random−
AIFB	1	0.0176	0.0129	0.7145	0.6767
AIFB	2	0.0162	0.0066	0.7386	0.7556
AIFB	5	0.1079	0.0520	0.7160	0.6085
AIFB	10	0.1829	0.0735	0.6338	0.6184
MUTAG	1	0.0789	−0.0013	0.5397	0.3151
MUTAG	2	0.1224	0.0205	0.3068	0.2744
MUTAG	5	0.1627	0.0475	0.1987	0.3352
MUTAG	10	0.0999	0.1592	0.2299	0.3462

36 explained nodes on AIFB, 68 on MUTAG.

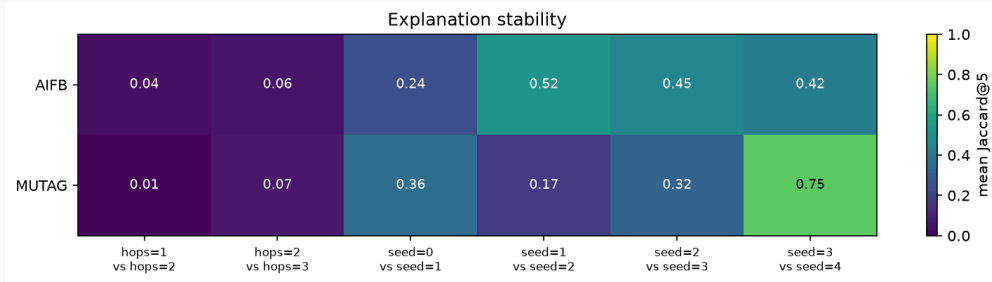
Sufficiency. On AIFB fid^- stays high at every k (≈ 0.7): no five features *suffice*, the model spreads evidence across the whole neighbourhood profile. On MUTAG the top-5 preserve the prediction markedly better than random-5 (0.199 vs. 0.335).

Fidelity+ vs. k against the random- k control



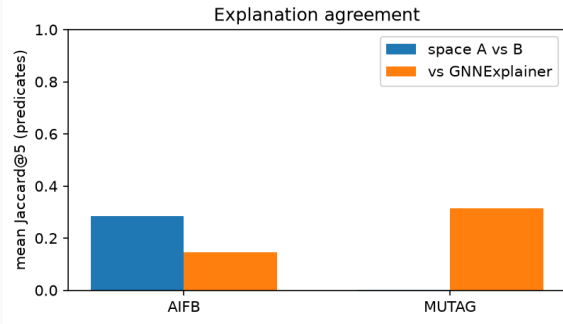
Interpretation. For $k \leq 5$ the GraphLIME curve sits clearly above the random control on both datasets: the explanations point at features the model actually uses. The inversion at $k = 10$ on MUTAG reflects the sweep, not the method — explanations carry fewer than ten non-zero β , so the top-10 set pads with irrelevant columns.

Explanation stability across seeds and hops



Interpretation. Across training seeds agreement is moderate (Jaccard@5 0.17–0.75): models of comparable accuracy use partially different features. Across hops it is near zero (0.01–0.07), because GraphLIME’s samples *are* the neighbourhood — **the radius is the single most consequential explanation hyperparameter**.

Feature-space sensitivity and baseline agreement



Comparison	Jaccard@5
AIFB, space A vs. B	0.2843
AIFB, vs. GNNExplainer	0.1456
MUTAG, space A vs. B	0.0029
MUTAG, vs. GNNExplainer	0.3153

Interpretation. A-vs-B overlap is moderate on AIFB and essentially zero on MUTAG, where space B shifts weight onto specific `rdf:type` objects (atom types) that space A cannot express. Agreement with GNNExplainer is limited on both — kernel dependence and a learned soft mask answer distinct questions — so the disagreement is itself the reported result.

Refusals and qualitative output

	Expl.	Ref.	Rate
AIFB	36	0	0.0000
MUTAG	68	0	0.0000

At 2 hops every test entity clears
`min_neighborhood = 10`. The guard was never
needed at this radius — not absent: it is exercised
by the test suite and by smaller-radius runs.

Explanations in the graph's vocabulary

The executed notebook prints sentence-form
explanations from the committed checkpoints, e.g. a
person correctly predicted into a research group *because*
of publication and author edges toward that group's
papers, or a MUTAG bond classified through its
atom-type context.

Refusals are printed inline, not skipped.

Limitations

- **Dependence, not causation.** β is a local, kernel-level account of dependence fitted on one neighbourhood; it does not license causal or global claims.
- **Fidelity on columns, not structure.** Masking a column removes the evidence of a predicate but leaves the edges intact; deleting the triples themselves would be more faithful to RDF semantics, at a substantially higher cost.
- **Neighbourhood distortions at the extremes.** Subsampled above 200 nodes, refused below 10 — and in between, the radius dominates the outcome.
- **Two small benchmarks.** 176 and 340 labelled entities put every mean on a few tens of test nodes; conclusions are claimed at this scale only.
- **The baseline is not like-for-like.** GNNExplainer is compared through its attribute mask, so the comparison covers feature attributions, not edge attributions.

Conclusions

1. GraphLIME cannot be applied to RDF as published: its input, a node feature matrix, **does not exist** in a triple store.
2. Defining that input from the graph's own predicates makes the method applicable *and* makes its output readable by a domain expert — explanations are statements about predicates, not tensor indices.
3. Letting the R-GCN consume the same matrix makes **fidelity measurable at all**, and the cost of interpretability is then visible: MUTAG above the one-hot baseline, AIFB below it.
4. The explanations beat the random control for $k \leq 5$ on both datasets, and the pipeline recovers a planted ground-truth rule.
5. The honest negative finding: on knowledge graphs of this size, **the neighbourhood radius dictates the explanation**, which bounds how far a single GraphLIME explanation should be trusted.

For the full argument, derivations and per-seed numbers, see `report/relazione.md`.

References

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